

Multiple Choice (10 marks)

1. Primary motor cortex activity results in
 - (a) bilateral contraction of antigravity limb muscles.
 - (b) bilateral contraction of limb musculature.
 - (c) contraction of ipsilateral limb musculature.
 - (d) contraction of contralateral limb musculature.
2. When asked where they would like to live for the rest of their lives most older adults say that they would prefer
 - (a) Moving to be close to their children
 - (b) Moving to Arizona or Florida if they could
 - (c) Residing in a very nice nursing home
 - (d) Staying in their own home
3. Which is a major characteristic of the bunyavirus family?
 - (a) Amplify by persistent infection of midges, mosquitoes, sand flies, ticks
 - (b) Predominantly mosquito borne
 - (c) Genetic recombination or reassortment
 - (d) Spread to humans via bites of mice and rats
4. Repeat core sequences consisting of 2, 3, or 4 base pairs are known as what?
 - (a) Single nucleotide polymorphisms (SNPs)
 - (b) Microsatellites
 - (c) Minisatellites
 - (d) Satellites
5. The stage of meiosis in which chromosomes pair and cross over is:
 - (a) prophase I
 - (b) metaphase I
 - (c) prophase II
 - (d) metaphase II

True / False (10 marks)

Answer True or False

6. True or False: Monosomy is the leading chromosome abnormality found in first trimester spontaneous miscarriages.
7. True or False: Many girls primarily learn about masturbation from reading literature on the subject.
8. True or False: Cytotoxic T cells can be activated by identifying virus peptides presented by MHC-I molecules on infected cells.
9. True or False: Herpes viruses can cause various types of cancer, including cervical and oral cancers.
10. True or False: Sterilization is the most common method of birth control used by married couples in the United States.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the role of proprioceptive information in the function of the temporomandibular joint, highlighting the contributions of the masseteric and auriculotemporal nerves.
12. Discuss the hormonal changes and physiological processes involved in ovulation, specifically identifying and analyzing the typical timing of ovulation within a 28-day menstrual cycle.

End of Paper